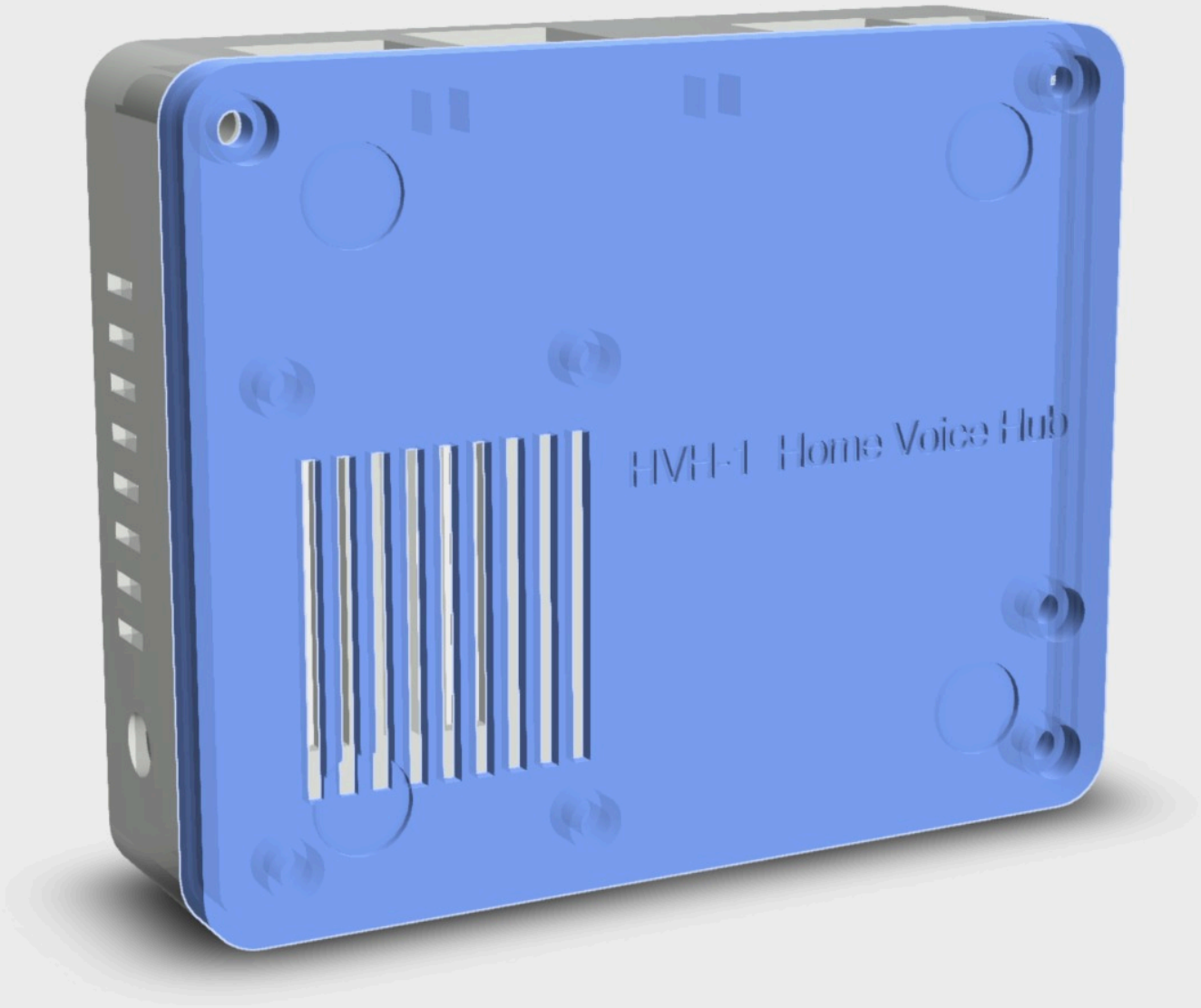
CASE STUDY

Claude in Embedded Engineering

HVH-1 Home Voice Hub: from a one-paragraph brief to a fully routed, DRC-clean, ready-to-order board, enclosure and OS image.

Raspberry Pi CM5 | 6-layer PCB | Zigbee / Thread | Home Assistant OS | 3D-printed enclosure

Sanjay Kumar · November 2026



THE BRIEF

One prompt, an end-to-end hardware product

What was asked

- Single-board computer for home automation, running Home Assistant OS
- Acts as a Claude-based voice assistant: listen, send to Claude, answer by text-to-speech
- Public SoC; USB-A for radios, on-board mic, speaker output
- 2.4 + 5 GHz Wi-Fi, Bluetooth, Zigbee (nice to have), USB-C power, microSD
- Circuit board, mechanical outline, 3D-printed enclosure (STL/3MF)
- SD image that pulls the latest HAOS
- "Keep working until it is fully sourceable and ready to manufacture"

WHAT CLAUDE DELIVERED

Schematic, footprints, 6-layer layout, Gerbers, JLCPCB BOM + placement file, STEP models, enclosure, a bootable SD image and a voice-assistant setup script. Everything is generated by scripts, so the design can be rebuilt.

132
schematic parts, 103 nets, 0 netlist errors

0
DRC errors and unconnected pads

111
placements, all top side, in-stock parts

6L
100 × 80 mm, impedance-controlled

2
printed parts: base + lid

18.5
latest HAOS baked into the SD image

HOW CLAUDE WORKED

The full embedded workflow, done as code

1 Requirements to SoC
Picked CM5 Lite Wireless: public, Wi-Fi 5 + BT on module, official HAOS support.

2 Parts and sourcing
Checked every part against JLC/LCSC stock data; DigiKey fallback for the Zigbee module.

3 Schematic
7 KiCad sheets generated from one design file; netlist cross-checked automatically.

4 Footprints
CM5 connectors, RJ45, USB3-A, mic and Zigbee module built and pin-checked against datasheets.

5 Placement and stack-up
Connector-edge layout, antenna keep-outs, 90/100 Ω differential pairs.

6 Routing and DRC
Custom grid router, rip-up and repair, iterated to 0 errors in KiCad DRC.

7 Mechanical
Parametric CadQuery enclosure driven by the same board coordinates.

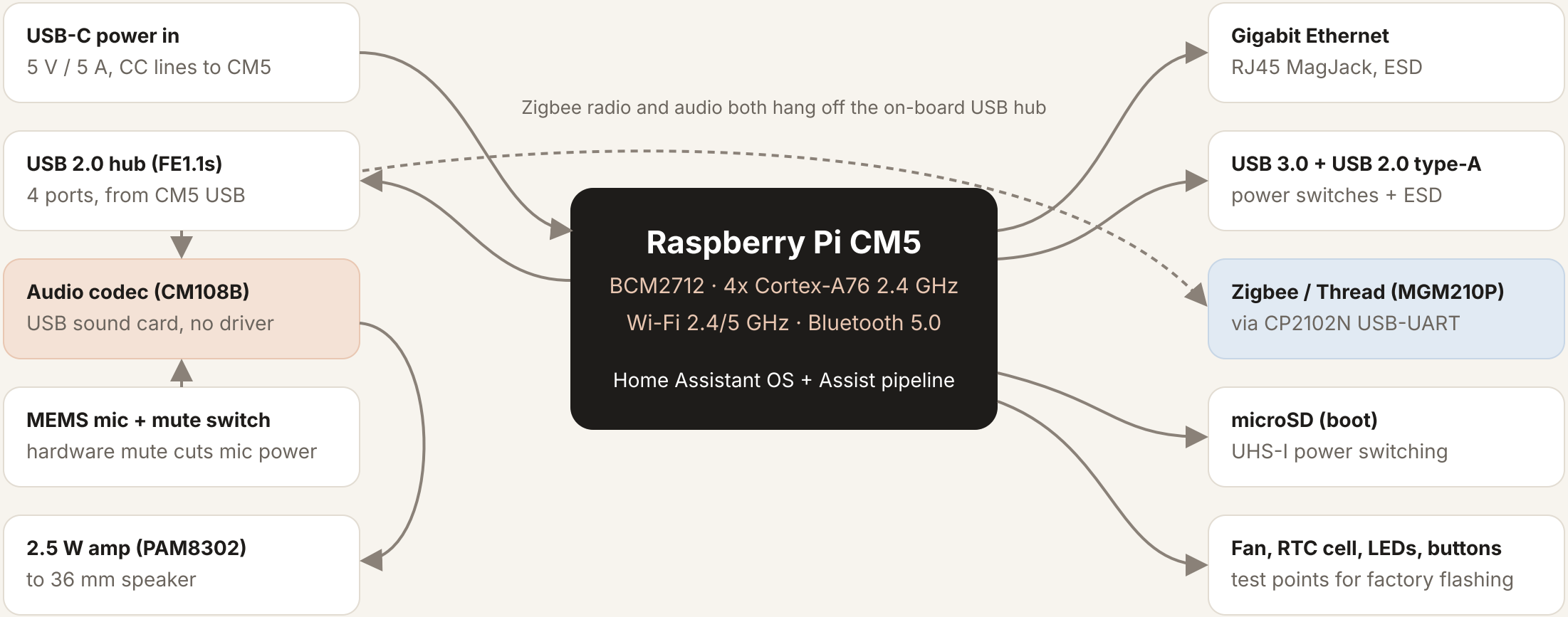
8 Software
HAOS image builder, hardware config, Claude voice pipeline provisioning.

WHY THIS MATTERS

Claude did not just answer questions about each step. It wrote and ran the tools, read the results (DRC reports, renders, netlists) and fixed the design in a loop, the way an engineer iterates in an EDA tool.

SYSTEM ARCHITECTURE

HVH-1 block diagram



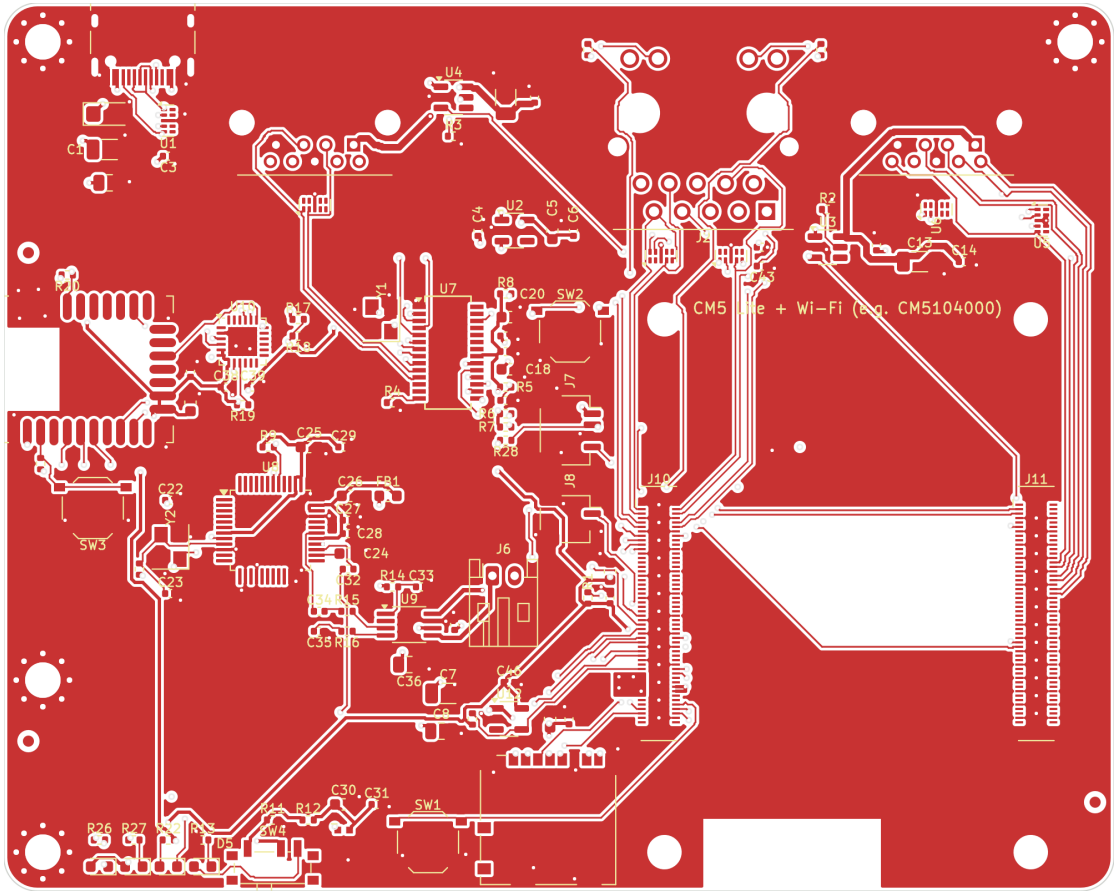
ENGINEERING JUDGEMENT

Decisions Claude made, and why

DECISION	CHOICE	REASONING
Compute	Raspberry Pi CM5 Lite Wireless (CM5104000)	Publicly available, dual-band Wi-Fi + BT on module, HAOS ships a CM5 build
Zigbee / Thread	Silicon Labs MGM210P, same as the HA Yellow	Reuses official, maintained Yellow radio firmware instead of a custom port
Audio	CM108B USB codec on an on-board hub	Shows up as a standard USB sound card in HAOS; no device-tree work needed
Power	USB-C CC pins wired straight to the CM5	Lets the CM5 negotiate 5 V / 5 A with the official 27 W supply
Board stack	4 layers abandoned for 6 (JLC06161H-3313)	The router proved the CM5 connector area could not be routed 100% on 2 signal layers
Signal integrity	90 Ω USB, 100 Ω Ethernet diff pairs, solid GND planes next to both outer layers	Calculated trace geometry from the stack-up; ordered with impedance control
Manufacturability	All parts top side, LCSC part numbers on every line	Single-side assembly keeps prototype cost down; stock checked at generation time

PCB LAYOUT

100 x 80 mm, 6 layers, fully routed



Top copper and silkscreen. CM5 sockets on the right, rear-edge connectors at the top, mic, LEDs and microSD on the front edge.

1,650
track segments

316
vias

0
DRC errors

0
unconnected pads

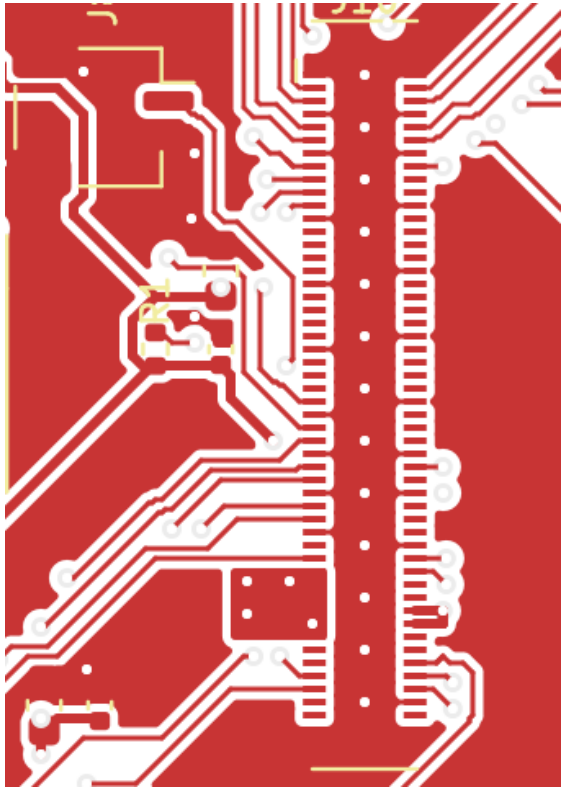
Stack: L1 signals · L2 GND · L3 low-speed signals · L4 +5 V · L5 GND · L6 signals

CLAUDE'S PART

Placed every part, wrote the router, read each DRC report and fixed the causes: rule mismatches, duplicate vias, blocked ground pins, crowded escape lanes.

UNDER THE HOOD

How the routing problem got solved



CM5 connector: ground channel with stitching vias between the 0.4 mm pitch pin rows.

1 **Own router when the standard tool was unavailable**

A* on a 0.1 mm grid with clearance maps, differential-pair coupling and via costs.

2 **Diagnosed why nets failed**

Rendered the board, found test points and capacitors blocking the microSD lane, and moved them.

3 **Planned connector escapes**

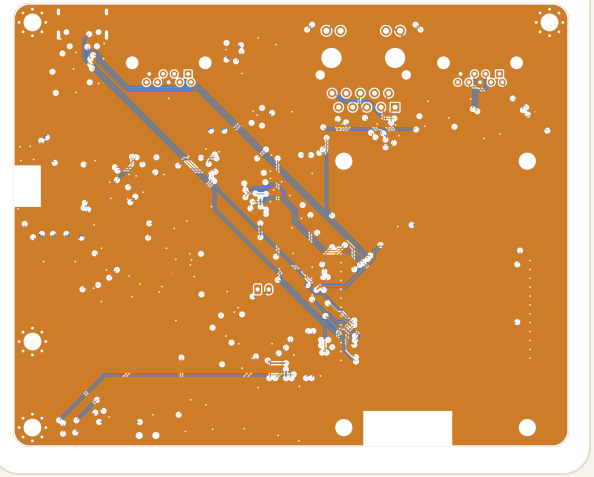
Searched stub-and-via positions for neighbouring CM5 pins together, so one escape no longer blocked the next.

4 **Rip-up and repair**

Kept a change only when the failure count dropped; fixed a hidden bug that reported partial routes as done.

5 **Closed out in KiCad DRC**

Matched netclass rules to the router, merged near vias, added +5 V stitching, filled pours.



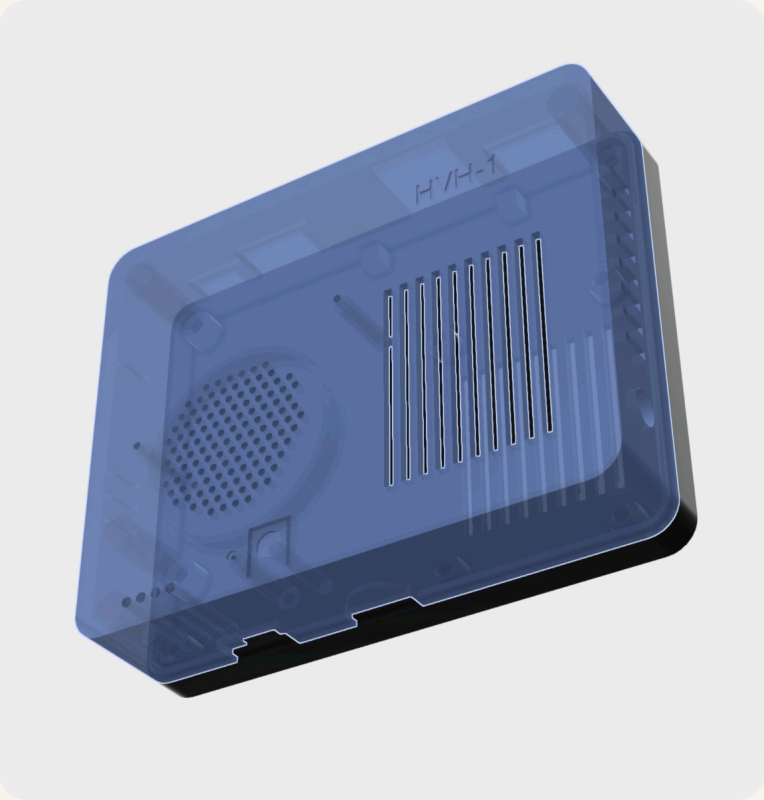
Bottom and inner signal layer (L3) used for crossings.

HONEST NOTE

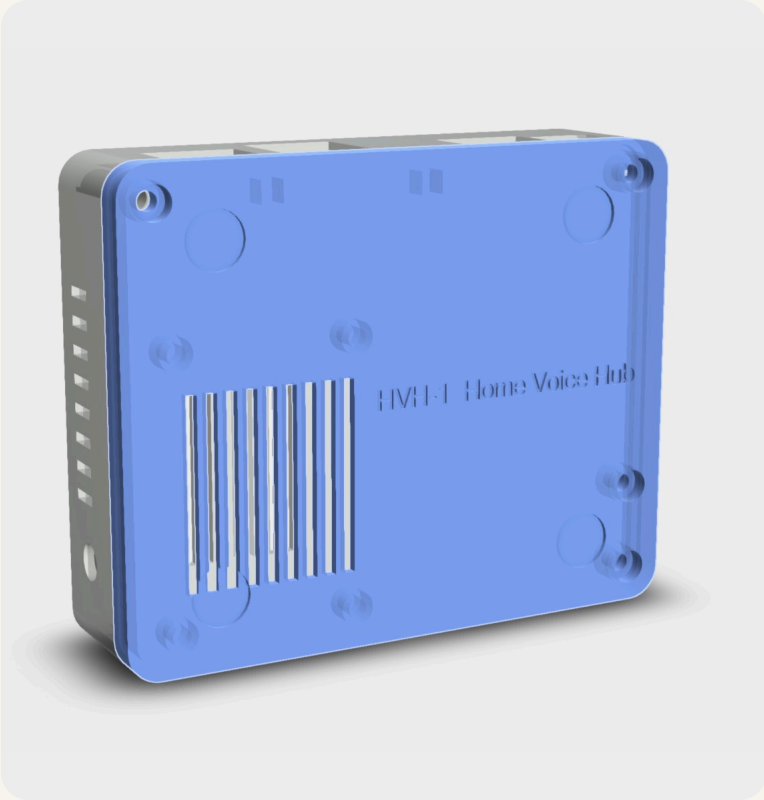
Routing was the slow part, about 10 hours of compute and iteration. Hand guidance on placement or a mature autorouter would cut this a lot.

MECHANICAL

Parametric enclosure from the same board data



Lid: speaker grille, CM5 vents, LED light pipes, mic port, power key.



Base: floor vents under the CM5, screw bosses, rubber-foot recesses.

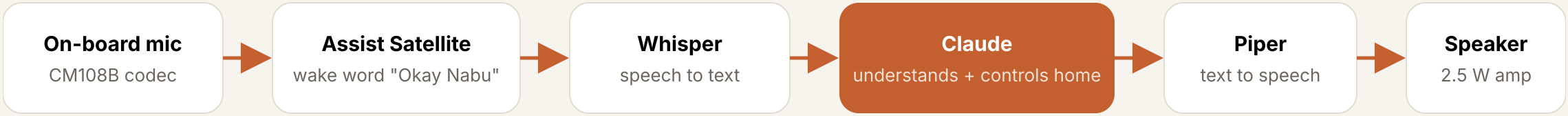
- Built in CadQuery, exported as STL, 3MF and STEP
- Cut-outs placed from the PCB file, so ports always line up
- Cantilever power key, sealed mic tube, SMA knock-out
- About 106 × 86 × 27 mm; prints without supports

CLAUDE'S PART

Read connector heights and courtyard outlines from the board, then designed every opening and boss around them.

SOFTWARE

From power-on to talking with Claude



SD-card image builder

- Finds and downloads the latest HAOS release, verifies it
- Writes HVH-1 settings: USB current, fan curve, Wi-Fi country, debug UART
- Adds Wi-Fi, SSH key and a stable Zigbee device name
- Built and checked here: HAOS 18.5, boot partition passes fsck

One-command assistant setup

- Installs Whisper, Piper and the Assist Satellite app
- Selects the on-board codec as mic and speaker
- Adds the Anthropic integration with your API key
- Creates the "HVH-1 Claude" pipeline and makes it the default

VERIFICATION

What was checked, and what still needs a human

Verified by Claude

- Schematic netlist matches the design file: 0 errors
- Every footprint pin matches its symbol
- KiCad DRC: 0 errors, 0 unconnected pads
- Parts in stock at JLC/LCSC when generated
- Enclosure solids valid; openings placed from board data
- SD image builds; boot partition passes a filesystem check

Needs engineering sign-off

- No prototype built or powered yet
- Independent EE review of schematic and layout
- Impedance confirmed by the fab on the real stack-up
- Zigbee module programmed over SWD at the factory
- Setup script not yet run against a live Home Assistant
- Print-fit check of the enclosure on the first board

TAKEAWAY

Claude compresses weeks of design work and states clearly where its checks stop. The engineer's role moves from drafting to reviewing, testing and signing off.

BUILD COST

Estimated cost for 2 units

ITEM	MADE IN CHINA (JLCPCB), SHIPPED TO INDIA	MADE IN INDIA
Bare boards (5) + assembly of 2	about ₹22,500 to ₹28,500	about ₹28,000 to ₹48,000
Indian customs on assembled boards	about ₹9,500 to ₹12,500	none
2 printed enclosures	₹1,500 to ₹3,500	₹1,200 to ₹1,800
Fab, assembly and enclosure	**about ₹34,000 to ₹45,000**	**about ₹31,000 to ₹50,000**
CM5 modules (2) and small hardware	about ₹15,000 to ₹16,000, either route	

Recommended: order boards from JLCPCB using the generated Gerber zip, BOM and placement file; print the enclosure locally. Budget about ₹50,000 to ₹62,000 for 2 complete units.

Estimates from November 2026 public price lists at about ₹94 per USD. Indian 6-layer prices are quote-only, so those figures are estimates; customs depends on how the courier classifies the boards.

LESSONS

Where Claude fits in embedded engineering

Where it accelerates most

- Part selection with live stock checks
- Schematic, BOM and pin-map generation
- Datasheet-driven footprints
- Mechanical design tied to the PCB
- OS images and provisioning scripts

Where it worked hardest

- Dense routing around a 0.4 mm pitch module
- Turning DRC reports into root-cause fixes
- Trading off layer count against cost

Best way to work with it

- Give clear constraints up front (layers, budget, fab)
- Review placement before routing starts
- Keep a human for bring-up, test and certification
- Keep designs as code, so every change can be regenerated

BOTTOM LINE

One brief produced a complete, manufacturable product package: board, enclosure and software. Claude acts as a tireless embedded co-engineer, and the human engineer stays in charge of review and validation.